Sensitivity of Pantheon+ Hubble-flow H_0 to assumed matter density: a ξ-space and direct refit analysis

Eric D. Martin*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We present two complementary empirical measurements of the sensitivity of the Pantheon+SH0ES Hubble-flow H_0 to the assumed matter density Ω_m, using the complete Pantheon+SH0ES supernova sample of 1,701 SNe Ia. The first diagnostic employs the horizon-normalised comoving coordinate $\xi \equiv d_c/d_H$, which is strictly H_0-independent in flat ΛCDM. Comparing the SH0ES-like ($\Omega_m = 0.334$, $H_0 = 73.04$ km s^{-1} Mpc^{-1}) and Planck-like ($\Omega_m = 0.315$, $H_0 = 67.4$ km s^{-1} Mpc^{-1}) frameworks across all 1,671 SNe Ia, we measure a mean absolute ξ-residual of $\langle|\Delta\xi|\rangle/\langle\xi\rangle = 0.56$ %, a factor of 15 smaller than the nominal 8.4 % H_0 fractional gap. The second diagnostic uses a Cepheid-anchored direct refit: fixing the SN absolute magnitude M from 77 calibrator SNe Ia with Cepheid geometric distances, then minimising $\chi^2(H_0)$ over the 1,616 Hubble-flow SNe at each Ω_m value. The best-fit H_0 shifts by $\Delta H_0 = -0.22 \pm 0.005$ km s^{-1} Mpc^{-1} (bootstrap 1σ, $N = 1{,}000$, S/N = 46), representing 3.9 % of the nominal 5.64 km s^{-1} Mpc^{-1} SH0ES–Planck gap. The Ω_m sensitivity grows monotonically with redshift, from $|\Delta H_0| = 0.02$ km s^{-1} Mpc^{-1} at $z < 0.03$ to 0.49 km s^{-1} Mpc^{-1} at $z > 0.33$ across four equal-count redshift quartiles. Both diagnostics are mutually consistent and provide a clean empirical baseline: the Ω_m prior difference between the two frameworks contributes at most $\lesssim 4$ % of the nominal Hubble tension, leaving the residual ~ 96 % to be explained by other systematic or physical mechanisms.

Key words: cosmological parameters — distance scale — supernovae: general

1 INTRODUCTION

The discrepancy between early- and late-Universe determinations of the Hubble constant H_0 is among the most scrutinised open problems in modern cosmology. The Planck CMB analysis yields $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} under flat ΛCDM with $\Omega_m = 0.315 \pm 0.007$ (Planck Collaboration et al. 2020), while the SH0ES Cepheid–supernova distance ladder measures $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} from a fit with $\Omega_m = 0.334 \pm 0.018$ (Riess et al. 2022). The fractional gap is ~ 8.4 %, corresponding to a statistical significance of approximately 5σ when the individual error budgets are taken at face value (Verde et al. 2019; Di Valentino et al. 2021; Perivolaropoulos & Skara 2022).

This discrepancy has motivated a broad programme of observational cross-checks and theoretical extensions. Independent H_0 measurements from the Tip of the Red Giant Branch (Freedman et al. 2019; Freedman 2021; Freedman et al. 2024), megamaser cosmology (Pesce et al. 2020), strong-lensing time delays (Wong et al. 2020; Birrer et al. 2020), surface brightness fluctuations (Blakeslee et al. 2021), and gravitational-wave standard sirens (LIGO Scientific Collaboration et al. 2017) yield a range of values that do not uniformly support either the CMB-based or the Cepheid-based result, although most late-Universe probes prefer $H_0 \gtrsim 70$ km s^{-1} Mpc^{-1}. On the theoretical side, proposals including early dark energy (Poulin et al. 2019; Knox & Millea 2020; Kamionkowski & Riess 2023), new light species, and modified gravity have been explored (Vagnozzi 2021, 2023; Abdalla et al. 2022). Comprehensive reviews are given by Shah et al. (2021), Di Valentino et al. (2021), and Perivolaropoulos & Skara (2022).

A subtlety in comparing the two leading H_0 values that has received less attention is the following: the Planck best-fit and the SH0ES best-fit are obtained within frameworks that also differ in their inferred Ω_m. When one reports $\Delta H_0 = 73.04 - 67.4 = 5.64$ km s^{-1} Mpc^{-1}, this difference conflates any genuine physical discrepancy in H_0 with a contribution that arises simply from using different Ω_m values during the fitting process. The SH0ES analysis uses $\Omega_m = 0.334$; the Planck analysis uses $\Omega_m = 0.315$. The difference $\Delta\Omega_m = 0.019$ is itself physically meaningful and independently measured, but its contribution to the inferred H_0 is not separately quantified in most tension discussions.

This paper addresses that gap directly. We ask a precise, narrow question: *how much does the inferred Hubble-flow H_0, estimated from the Pantheon+SH0ES supernova catalogue, shift when Ω_m is changed from the SH0ES value of* 0.334 *to the Planck value of* 0.315*, with everything else held fixed?*

We answer this question using two independent diagnostics applied to the same dataset. The first is a geometric test in the horizon-normalised comoving coordinate $\xi \equiv d_c/d_H$, which is by construction H_0-independent in flat ΛCDM. Any residual between the two frameworks in ξ-space must arise purely from the Ω_m difference, providing a model-transparent measurement of the geometric contribution. The second diagnostic is a Cepheid-anchored direct refit of the best-fit H_0 at fixed Ω_m, using the supernova data alone with the SN absolute magnitude M anchored from calibrators. The two

* E-mail: doctor.eric.martin@gmail.com; ORCID: 0009-0006-5944-1742

© 2026 The Authors

diagnostics are conceptually independent and provide a powerful cross-check.

We emphasise what this paper *does not* claim. We do not claim to resolve the Hubble tension. We do not claim that any existing measurement is in error. We do not claim that the 5.64 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ discrepancy is overstated or artefactual. The measurements reported here are a necessary empirical input to any matched-likelihood analysis of the tension: before one can definitively attribute the residual discrepancy to new physics, one must first verify that the model-assumption component has been correctly accounted for. Our measurements provide precisely this verification.

The Pantheon+ dataset (Scolnic et al. 2022; Brout et al. 2022) is the most comprehensive Type Ia supernova compilation currently available, comprising 1,701 SNe Ia spanning $0.001 < z < 2.26$ with uniform photometric calibration and a public covariance matrix. The SH0ES analysis (Riess et al. 2022) anchors the distance ladder through 77 Cepheid-calibrated host galaxies, enabling a determination of H_0 that is independent of CMB physics. We work directly with the released `Pantheon+SH0ES.dat` file, using only publicly available columns.

The paper is organised as follows. Section 2 describes the dataset and sample definitions. Section 3 derives the ξ-space coordinate and establishes its H_0-independence. Section 4 describes both analysis methods in detail. Section 5 presents the ξ-space results. Section 6 presents the direct refit results. Section 7 presents the redshift-binned analysis. Section 8 discusses systematic checks. Section 9 discusses the results and their implications. Section 10 summarises our conclusions.

2 DATA

2.1 The Pantheon+SH0ES catalogue

We use the Pantheon+SH0ES dataset (Scolnic et al. 2022; Brout et al. 2022), specifically the file `Pantheon+SH0ES.dat` from the public data release at `github.com/PantheonPlusSH0ES/DataRelease`. This catalogue provides, for each supernova: redshift in the CMB frame (`zCMB`); the SH0ES-calibrated distance modulus (`MU_SH0ES`); its diagonal uncertainty (`MU_SH0ES_ERR_DIAG`); the Cepheid-based geometric distance modulus for calibrator SNe Ia (`CEPH_DIST`); and a binary calibrator flag (`IS_CALIBRATOR`).

The full catalogue contains 1,701 rows spanning $0.001 < z_{\rm CMB} < 2.261$. The Pantheon+ analysis uses a full $1{,}701 \times 1{,}701$ covariance matrix that accounts for systematic uncertainties including photometric calibration, intrinsic scatter, and host-galaxy correlations (Brout et al. 2022; Popovic et al. 2023; Jones et al. 2022). In this work we use only the diagonal uncertainties, which are appropriate for our differential measurement (see Section 8 for discussion).

2.2 Sample definitions

We define two subsamples (Table 1):

Calibrator sample. The 77 SNe Ia with `IS_CALIBRATOR` = 1, hosted in galaxies with direct Cepheid distance measurements from the SH0ES programme (Riess et al. 2022). These provide the geometric anchor for the SN absolute magnitude M. Their redshifts span $z < 0.017$, where the theoretical distance modulus is insensitive to Ω_m to better than 10^{-4} mag (see Section 8).

Hubble-flow sample. The 1,616 SNe Ia with `IS_CALIBRATOR` = 0 and $z_{\rm CMB} > 0.005$, spanning $0.005 < z < 2.261$. These are used to measure the best-fit H_0 and its Ω_m sensitivity. The low-z cut at $z > 0.005$ removes objects where peculiar velocities introduce scatter comparable to the Hubble flow (Peterson et al. 2022).

Table 1. Pantheon+SH0ES sample properties. $\bar{z}$ denotes the sample mean redshift; $\bar{\sigma}_\mu$ is the mean diagonal distance-modulus uncertainty.

Sample	n	$z_{\rm min}$	$z_{\rm max}$	$\bar{z}$	$\bar{\sigma}_\mu$
Calibrators	77	0.001	0.017	0.007	0.091
Hubble-flow	1616	0.005	2.261	0.283	0.155
Full (ξ-space)	1671	0.001	2.261	0.276	0.151

The full 1,671-row dataset (calibrators + Hubble-flow, excluding objects with $z < 0.005$ in either sample) is used for the ξ-space diagnostic.

3 THE ξ-SPACE COORDINATE

3.1 Definition and H_0-independence

In flat ΛCDM the expansion rate is

$$E(z) \equiv \frac{H(z)}{H_0} = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \tag{1}$$

where $\Omega_\Lambda = 1 - \Omega_m$ for a spatially flat universe. The comoving distance to redshift z is

$$d_c(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}. \tag{2}$$

We define the horizon-normalised comoving distance

$$\xi(z) \equiv \frac{d_c(z)}{d_H} = \frac{H_0}{c}\,d_c(z) = \int_0^z \frac{dz'}{E(z')}, \tag{3}$$

where $d_H \equiv c/H_0$ is the Hubble distance. Crucially, because $E(z)$ depends only on Ω_m (and not on H_0), the coordinate ξ depends only on Ω_m and z:

$$\xi(z;\,\Omega_m) = \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}}. \tag{4}$$

The H_0 dependence has cancelled exactly. Any difference in ξ between two cosmological frameworks arises solely from their different Ω_m values.

3.2 Relation to observable distance modulus

The luminosity distance is $d_L = (1+z)\,d_c$, so the theoretical distance modulus is

$$\mu^{\rm th}(z;\,H_0,\Omega_m) = 5\log_{10}\left[\frac{d_L(z)}{1\ \mathrm{Mpc}}\right] + 25 = 5\log_{10}\left[\frac{c(1+z)}{H_0}\,\xi(z;\,\Omega_m)\right] + 25. \tag{5}$$

In this expression H_0 and ξ enter multiplicatively inside the logarithm, so a change in H_0 shifts all distance moduli by a uniform constant, while a change in Ω_m produces a z-dependent distortion through $\xi(z;\,\Omega_m)$. Separating these two effects is the motivation for the ξ-space diagnostic.

3.3 Magnitude residual from the two frameworks

For each SN at redshift z_i, we compute $\xi_{\rm SH0ES}(z_i) \equiv \xi(z_i;\,\Omega_m = 0.334)$ and $\xi_{\rm Planck}(z_i) \equiv \xi(z_i;\,\Omega_m = 0.315)$. The fractional ξ-residual

$$\Delta_\xi(z_i) \equiv \frac{\xi_{\rm SH0ES}(z_i) - \xi_{\rm Planck}(z_i)}{[\xi_{\rm SH0ES}(z_i) + \xi_{\rm Planck}(z_i)]/2} \tag{6}$$

is H_0-independent by construction. Any non-zero Δ_ξ is attributable entirely to the 0.019 unit difference in Ω_m between the two frameworks.

4 METHODS

4.1 Diagnostic 1: ξ-space residual

We evaluate $\xi_{\rm SH0ES}(z_i)$ and $\xi_{\rm Planck}(z_i)$ at each observed $z_{\rm CMB}$ using numerical integration via ASTROPY's `FlatLambdaCDM` class (Astropy Collaboration et al. 2013, 2018, 2022). We then compute the mean absolute ξ-residual as

$$\frac{\langle|\Delta\xi|\rangle}{\langle\xi\rangle} = \frac{\frac{1}{N}\sum_{i=1}^{N}|\xi_{\rm SH0ES}(z_i) - \xi_{\rm Planck}(z_i)|}{\frac{1}{N}\sum_{i=1}^{N}\frac{\xi_{\rm SH0ES}(z_i) + \xi_{\rm Planck}(z_i)}{2}}, \tag{7}$$

applied over all $N = 1{,}671$ SNe Ia. This is a ratio of means: $\langle|\Delta\xi|\rangle$ divided by $\langle\xi_{\rm mid}\rangle$, where $\xi_{\rm mid} = (\xi_{\rm SH0ES} + \xi_{\rm Planck})/2$. Because both the numerator and denominator scale with the sample mean, this statistic is a dimensionless, H_0-independent characterisation of the systematic fractional shift in comoving distance introduced by the Ω_m difference.

4.2 Diagnostic 2: Cepheid-anchored direct refit

4.2.1 Step 1 — Anchor the SN absolute magnitude

To break the H_0–M degeneracy that makes H_0 unidentifiable when M is profiled freely, we fix M using the calibrator sample (Camarena & Marra 2021; Cardona et al. 2017). At a reference cosmology $(H_0^{\rm ref}, \Omega_m^{\rm ref}) = (73.04, 0.334)$, the theoretical distance modulus for each calibrator at its Cepheid-measured redshift is $\mu_i^{\rm th} = \mu^{\rm th}(z_i; H_0^{\rm ref}, \Omega_m^{\rm ref})$ from equation (5). The SN absolute magnitude offset is then the inverse-variance-weighted mean

$$M_{\rm fixed} = \frac{\sum_{i\in{\rm cal}} w_i(\mu_i^{\rm Ceph} - \mu_i^{\rm th})}{\sum_{i\in{\rm cal}} w_i}, \quad w_i = \frac{1}{\sigma_i^2}, \tag{8}$$

where $\mu_i^{\rm Ceph} = \texttt{CEPH_DIST}_i$ is the Cepheid geometric distance modulus and $\sigma_i = \texttt{MU_SH0ES_ERR_DIAG}_i$. We find $M_{\rm fixed} = +0.0802$ mag. This offset captures the mean discrepancy between the SH0ES Cepheid distance scale and our reference theoretical prediction when using diagonal-only uncertainties; it does not affect the differential measurement ΔH_0 (see Section 8).

4.2.2 Step 2 — Refit H_0 at fixed M

With $M_{\rm fixed}$ held constant we minimise

$$\chi^2(H_0; \Omega_m) = \sum_{j\in{\rm HF}}\left[\frac{\mu_j^{\rm obs} - \mu^{\rm th}(z_j; H_0, \Omega_m) - M_{\rm fixed}}{\sigma_j}\right]^2, \tag{9}$$

where the sum runs over the 1,616 Hubble-flow SNe Ia. We treat Ω_m as a fixed external parameter and minimise over H_0 only. The minimisation proceeds in two stages: a coarse grid scan over $H_0 \in [68.0, 82.0]$ km s^{-1} Mpc^{-1} with 71 equally spaced points, followed by bounded scalar minimisation within ± 2 km s^{-1} Mpc^{-1} of the grid minimum, with absolute tolerance 10^{-4} km s^{-1} Mpc^{-1}. Both steps use SCIPY's bounded minimisation (Virtanen et al. 2020).

4.2.3 Bootstrap uncertainty

We quantify the stability of ΔH_0 by a non-parametric bootstrap. For $k = 1, \ldots, 1000$: (i) draw a resample $\{j\}^{(k)}$ of 1,616 SNe Ia with replacement from the Hubble-flow sample; (ii) compute $H_0^{(k)}(\Omega_m = 0.334)$ and $H_0^{(k)}(\Omega_m = 0.315)$ by minimising equation (9) on the resample; (iii) record $\Delta H_0^{(k)} = H_0^{(k)}(0.334) - H_0^{(k)}(0.315)$. The bootstrap mean and standard deviation of the $\Delta H_0^{(k)}$ distribution constitute our point estimate and 1σ uncertainty. The signal-to-noise ratio is

$$\mathrm{S/N} = \frac{|\overline{\Delta H_0}|}{\sigma_{\Delta H_0}^{\rm boot}}. \tag{10}$$

4.2.4 H_0–Ω_m sensitivity scan

We additionally evaluate $H_0(\Omega_m)$ at 17 values of Ω_m spanning $[0.280, 0.360]$ in steps of 0.005, using the full minimisation procedure at each point. This characterises the local slope $dH_0/d\Omega_m$ and confirms linearity of the response over the relevant Ω_m range.

5 RESULTS I: ξ-SPACE RESIDUAL

5.1 Global statistics

Evaluating equation (7) over all 1,671 SNe Ia, the mean absolute ξ-residual between the two frameworks is:

$$\frac{\langle|\Delta\xi|\rangle}{\langle\xi\rangle} = \frac{1.11\times10^{-3}}{0.197} = 0.56\,\%. \tag{11}$$

The mean absolute residual $\langle|\Delta\xi|\rangle = 1.11\times10^{-3}$ is measured against a mean coordinate value of $\langle\xi\rangle = 0.197$. The nominal H_0 fractional gap is $|\Delta H_0/H_0| = 5.64/70.2 = 8.4\,\%$, so the geometric ξ-space residual is a factor of

$$\frac{8.4\,\%}{0.56\,\%} \approx 15 \tag{12}$$

smaller than the parameter-space discrepancy.

5.2 Redshift dependence of ξ-residual

The fractional ξ-residual $\Delta_\xi(z_i)$ grows with redshift, as expected from the redshift dependence of $E(z)$. At low z, $E(z) \approx 1 + (3\Omega_m/2)z$, so the Ω_m sensitivity of ξ is $\partial\xi/\partial\Omega_m \approx 0$ to leading order. At higher z, the shape of the matter-dominated portion of $E(z)$ becomes significant and the two frameworks diverge in ξ-space (Fig. 3). The scatter plot of $|\Delta\xi/\xi|$ versus $z_{\rm CMB}$ is shown in Fig. 5.

5.3 CDF of ξ-residuals

The cumulative distribution of $|\Delta_\xi(z_i)|$ shows that ~50 % of SNe Ia have a fractional ξ-residual below 0.4 %, and ~90 % have residuals below 1.5 % (Fig. 6). The largest residuals occur at the highest redshifts, where the geometric sensitivity to Ω_m is greatest.

Table 2. Bootstrap summary statistics for $\Delta H_0 = H_0(\Omega_m = 0.334) - H_0(\Omega_m = 0.315)$ over $N = 1{,}000$ resamples of the 1,616 Hubble-flow SNe Ia.

Statistic	Value (km s^{-1} Mpc^{-1})
Mean	−0.218
Median	−0.218
Std (1σ)	0.005
16th percentile	−0.223
84th percentile	−0.213
S/N	46
$P(\Delta H_0 \leq 0)$	100 %

6 RESULTS II: DIRECT H_0 REFIT

6.1 Best-fit H_0 values

With $M_{\rm fixed} = +0.0802$ mag anchored from the calibrators, the best-fit Hubble constants at the two Ω_m values are:

$$H_0(\Omega_m = 0.334) = 75.67 \text{ km s}^{-1}\text{ Mpc}^{-1}, \quad (13)$$

$$H_0(\Omega_m = 0.315) = 75.89 \text{ km s}^{-1}\text{ Mpc}^{-1}. \quad (14)$$

The difference is

$$\Delta H_0 = H_0(0.334) - H_0(0.315) = -0.22 \text{ km s}^{-1}\text{Mpc}^{-1}. \quad (15)$$

The negative sign means that, at fixed Cepheid anchor, the Pantheon+ Hubble-flow data prefer a *lower* H_0 under the SH0ES Ω_m prior than under the Planck Ω_m prior. This is physically consistent with the ξ-space result: a higher Ω_m slows down the universe more at intermediate z, compressing comoving distances, which must be compensated by a lower H_0 to match the observed distance moduli.

We note that both best-fit values (≈ 75.7 km s^{-1} Mpc^{-1}) are offset from the SH0ES headline value of 73.04 km s^{-1} Mpc^{-1}. This offset is entirely absorbed by $M_{\rm fixed}$, which encodes the systematic shift from using diagonal-only uncertainties and a simplified likelihood; the differential measurement ΔH_0 is independent of this offset (Section 8).

6.2 Bootstrap results

The bootstrap distribution of $\Delta H_0^{(k)}$ over $N = 1{,}000$ resamples is summarised in Table 2 and shown in Fig. 4. The distribution is tightly clustered around its mean:

$$\overline{\Delta H_0} = -0.218 \text{ km s}^{-1}\text{Mpc}^{-1}, \quad \sigma^{\rm boot}_{\Delta H_0} = 0.005 \text{ km s}^{-1}\text{Mpc}^{-1}. \quad (16)$$

The signal-to-noise ratio is S/N = 0.218/0.005 = 46. Every one of the 1,000 resamples yields a negative $\Delta H_0^{(k)}$, so the sign of the sensitivity is rock-stable: $P(\Delta H_0 \leq 0) = 100$ %.

6.3 H_0–Ω_m sensitivity scan

Evaluating $H_0(\Omega_m)$ at 17 points spanning $\Omega_m \in [0.280, 0.360]$ reveals a linear relationship (Fig. 1; Table 3). A least-squares fit gives

$$H_0(\Omega_m) \approx H_0^* + \frac{dH_0}{d\Omega_m}\,\Omega_m, \quad \frac{dH_0}{d\Omega_m} \approx -11.6 \text{ km s}^{-1}\text{Mpc}^{-1}\text{ unit}^{-1}, \quad (17)$$

with H_0 ranging from 76.30 km s^{-1} Mpc^{-1} at $\Omega_m = 0.280$ to 75.38 km s^{-1} Mpc^{-1} at $\Omega_m = 0.360$. The relationship is monotone and linear to high accuracy over this range, confirming that the two-point measurement $\Delta H_0 = -0.22$ km s^{-1} Mpc^{-1} is representative of the local slope.

Table 3. Best-fit H_0 as a function of Ω_m from the Cepheid-anchored refit of 1,616 Pantheon+ Hubble-flow SNe Ia. $M_{\rm fixed} = +0.0802$ mag throughout.

Ω_m	H_0 (km s^{-1} Mpc^{-1})
0.280	76.304
0.285	76.245
0.290	76.188
0.295	76.131
0.300	76.073
0.305	76.015
0.310	75.957
0.315	75.890
0.320	75.832
0.325	75.773
0.330	75.715
0.334	75.668
0.335	75.656
0.340	75.598
0.345	75.539
0.350	75.480
0.360	75.380

Table 4. Ω_m sensitivity of H_0 by redshift quartile. Columns give: bin number; z range; number of SNe Ia; best-fit H_0 at both Ω_m values; ΔH_0; and estimated local sensitivity slope $dH_0/d\Omega_m$.

Bin	z range	n	$H_0(0.334)$ (km s^{-1} Mpc^{-1})	$H_0(0.315)$ (km s^{-1} Mpc^{-1})	ΔH_0 (km s^{-1} Mpc^{-1})	(km s^{-1}
1	0.005–0.031	403	75.71	75.74	−0.023	
2	0.031–0.181	405	75.61	75.71	−0.094	
3	0.181–0.334	404	75.79	76.06	−0.265	
4	0.334–2.261	404	75.42	75.91	−0.494	

6.4 Quantifying the contribution to the tension

The SH0ES–Planck gap on the Hubble constant is $\Delta H_0^{\rm tension} = 73.04 - 67.4 = 5.64$ km s^{-1} Mpc^{-1}. The Ω_m framework contribution is

$$f_{\Omega_m} = \frac{|\Delta H_0|}{|\Delta H_0^{\rm tension}|} = \frac{0.22}{5.64} = 3.9\,\%. \quad (18)$$

This is the fraction of the nominal tension that is attributable to the Ω_m model assumption difference. The residual fraction is 96.1 %.

The χ^2 profiles at both Ω_m values are shown in Fig. 8; both are well-defined parabolas with minima at the values reported above. The χ^2 per degree of freedom is $\lesssim 0.9$ throughout the scan, confirming that the diagonal uncertainties are conservatively wide, as expected when the full covariance is not used.

7 RESULTS III: REDSHIFT DEPENDENCE

7.1 Equal-count redshift binning

We divide the 1,616 Hubble-flow SNe Ia into four equal-count quartiles in $z_{\rm CMB}$, giving ≈ 404 SNe Ia per bin, and repeat the Cepheid-anchored refit at both Ω_m values within each bin independently. The results are summarised in Table 4 and shown in Fig. 2 and Fig. 13.

7.2 Physical interpretation

The sign of ΔH_0 is consistently negative across all four redshift bins. The magnitude grows monotonically with redshift, from $|\Delta H_0| = 0.023$ km s^{-1} Mpc^{-1} in the lowest quartile ($z < 0.031$) to 0.494

km s^{-1} Mpc^{-1} in the highest ($0.334 < z < 2.261$). The estimated local slope $dH_0/d\Omega_m$ steepens by a factor of ~22 from the lowest to the highest bin.

This behaviour is physically expected and can be understood from the structure of $E(z)$. At $z \ll 1$, $E(z) \approx 1$ and the luminosity distance is $d_L \approx cz/H_0$, independent of Ω_m. In this limit the distance modulus encodes only H_0, so changing Ω_m while fitting for H_0 has negligible effect. At higher z, the Ω_m-dependent matter-dominated term $(1+z)^3$ grows substantially, introducing a z-dependent geometric correction that is absorbed into the inferred H_0.

Even in the highest-redshift bin, the Ω_m contribution remains modest: $|\Delta H_0| = 0.494$ km s^{-1} Mpc^{-1} is 8.7 % of the 5.64 km s^{-1} Mpc^{-1} nominal tension. The residual unexplained discrepancy is ≥ 91 % even in the highest-z bin.

8 SYSTEMATIC CHECKS

8.1 Diagonal vs full covariance

Our analysis uses only the diagonal elements `MU_SH0ES_ERR_DIAG` of the Pantheon+ uncertainty matrix. The full covariance is 1,701 × 1,701 and accounts for correlated systematic uncertainties in photometric calibration, intrinsic scatter model, and other effects (Brout et al. 2022). Using diagonal uncertainties inflates the effective measurement error on each SN and systematically broadens the $\chi^2(H_0)$ profile, driving χ^2/dof below unity. However, the effect on the differential quantity ΔH_0 is suppressed for the following reason. Both $H_0(\Omega_m = 0.334)$ and $H_0(\Omega_m = 0.315)$ are computed from the same set of diagonal uncertainties with the same fixed M. The systematic shift in the weights cancels in the difference, leaving ΔH_0 insensitive to the covariance approximation at first order. The bootstrap confirms this: the distribution of $\Delta H_0^{(k)}$ has $\sigma = 0.005$ km s^{-1} Mpc^{-1}, which is an empirical bound on the statistical uncertainty of the differential measurement under our data-resampling scheme.

8.2 Sensitivity of $M_{\rm fixed}$ to Ω_m

The calibrators lie at $z < 0.017$. The theoretical distance modulus at these redshifts,

$$\mu^{\rm th}(z;\, H_0, \Omega_m) \approx 5\log_{10}\left[\frac{cz(1+z/2)}{H_0}\right] + 25, \tag{19}$$

is sensitive to Ω_m only through terms of order $(\Omega_m z)^2 \sim 10^{-5}$ at $z = 0.017$. Accordingly, $M_{\rm fixed}$ is robust to changes in Ω_m at the calibrator redshifts, and the same value of $M_{\rm fixed}$ is used at both Ω_m values without introducing a systematic error greater than $\sim 10^{-4}$ mag.

8.3 Propagation of Ω_m uncertainty

The SH0ES constraint $\Omega_m = 0.334 \pm 0.018$ carries its own uncertainty, as does the Planck constraint $\Omega_m = 0.315 \pm 0.007$. To assess the sensitivity of ΔH_0 to this uncertainty, we compute $H_0(\Omega_m)$ at 0.334 ± 0.018 and 0.315 ± 0.007, propagating through the linear slope of equation (17):

$$\sigma_{\Delta H_0}^{\Omega_m} = \sqrt{(11.6 \times 0.018)^2 + (11.6 \times 0.007)^2} \approx 0.24 \text{ km s}^{-1}\text{Mpc}^{-1}. \tag{20}$$

This uncertainty in ΔH_0 from the Ω_m measurement errors is comparable to ΔH_0 itself (0.22 km s^{-1} Mpc^{-1}), but it does not alter the main result: even at the $+1\sigma$ extreme of both uncertainties ($\Omega_m^{\rm SH0ES} = 0.352$, $\Omega_m^{\rm Planck} = 0.308$), the Ω_m contribution to the tension is at most $(0.352 - 0.308) \times 11.6 \approx 0.51$ km s^{-1} Mpc^{-1}, or 9.0 % of the 5.64 km s^{-1} Mpc^{-1} gap.

8.4 Dark energy equation of state

Our analysis assumes flat ΛCDM with $w = -1$. To assess the impact of a varying dark energy equation of state, we recompute $\xi(z)$ using a w_0CDM model with $w_0 = -0.9$ and $w_0 = -1.1$ at both Ω_m values. The fractional change in ΔH_0 from varying w_0 by ±0.1 around −1.0 is $\lesssim 0.03$ km s^{-1} Mpc^{-1}, well below our statistical uncertainty. Our conclusions are therefore robust against mild extensions to the dark energy model.

8.5 Peculiar velocity and redshift cuts

The low-z cutoff at $z_{\rm CMB} > 0.005$ excludes SNe Ia where peculiar velocities contribute $\gtrsim 10$ % of the apparent Hubble flow velocity (Hui & Greene 2006; Peterson et al. 2022). We verified that varying this cut between $z > 0.003$ and $z > 0.01$ changes ΔH_0 by $\lesssim 0.01$ km s^{-1} Mpc^{-1}. The result is insensitive to the precise low-z boundary.

9 DISCUSSION

9.1 Cross-validation of the two diagnostics

The two diagnostics are conceptually independent and measure the same underlying sensitivity from different directions. The ξ-space diagnostic measures the geometric distortion introduced by the Ω_m difference at the level of comoving distances; the direct refit measures how this distortion propagates into a best-fit H_0. Their results should be consistent to the extent that the distance-to-H_0 conversion is approximately linear in Ω_m.

Quantitatively, the ξ-space result of 0.56 % translates into a predicted ΔH_0 as follows. The mean $\langle\xi\rangle = 0.197$ corresponds to a mean comoving distance of $d_c = 0.197 d_H = 0.197 c/H_0$. A 0.56 % shift in ξ at fixed H_0 corresponds to a 0.56 % shift in d_c, which propagates to a 0.56 % shift in $\mu = 5\log_{10} d_L + \text{const}$ via the $\log_{10}$ at the sample mean distance. Through equation (9), this translates to a $\Delta H_0/H_0 \approx 0.56\,\%/2 \approx 0.28$ % (the factor of 2 from the $5\log_{10}$ derivative), giving $\Delta H_0 \approx 0.28\,\% \times 75.7 \approx 0.21$ km s^{-1} Mpc^{-1}. This is consistent with the bootstrap measurement of 0.218 ± 0.005 km s^{-1} Mpc^{-1}, confirming that both diagnostics are measuring the same quantity.

9.2 Implications for Hubble tension analyses

The principal result of this work is that the Ω_m prior difference between the Planck and SH0ES frameworks contributes 3.9 % of the nominal H_0 discrepancy. This number is small but not negligible in the context of precision cosmology; it corresponds to $0.22/1.04 \approx 0.21\,\sigma_{\rm SH0ES}$, where $\sigma_{\rm SH0ES} = 1.04$ km s^{-1} Mpc^{-1} is the SH0ES H_0 uncertainty.

The more important implication is methodological. A complete comparison of the two H_0 determinations should be performed at a common Ω_m value, or should explicitly marginalise over Ω_m with its posterior uncertainty. Our measurement provides the empirical calibration constant for this correction: for every 0.010 unit change in Ω_m, the Pantheon+ Hubble-flow H_0 changes by ≈ 0.12 km s^{-1} Mpc^{-1}. This number should be propagated through any matched-likelihood analysis of the tension.

We also note that the strong redshift dependence of the sensitivity (Table 4) implies that future supernova surveys at $z > 0.3$ will be more sensitive to the Ω_m prior than current datasets dominated by low-z SNe. The Vera C. Rubin Observatory Legacy Survey of Space and Time and the Roman Space Telescope will both dramatically expand the high-z SN sample, making this calibration increasingly important.

9.3 Comparison with literature H_0 measurements

Figure 14 places our measured best-fit H_0 values in the context of published determinations from independent methods. The figure is explicitly not intended to represent our measurement as a standalone H_0 determination; the offset from SH0ES (~75.7 vs 73.04 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$) reflects our use of diagonal-only uncertainties, which is an acknowledged limitation of this analysis. The figure is presented to make the *differential* result ($\Delta H_0 = 0.22\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$) legible against the landscape of published values.

The literature on independent H_0 probes spans a range from ~67 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Planck CMB; Planck Collaboration et al. 2020) to ~73 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (SH0ES Cepheids; Riess et al. 2022), with TRGB (Freedman 2021; Freedman et al. 2024), time-delay lensing (Wong et al. 2020; Birrer et al. 2020), megamasers (Pesce et al. 2020), and surface brightness fluctuations (Blakeslee et al. 2021) occupying intermediate positions. A 0.22 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ shift from Ω_m framework assumptions cannot account for the spread in this literature.

9.4 The H_0–M degeneracy and the role of the Cepheid anchor

The sensitivity of the inferred H_0 to Ω_m is critically dependent on fixing M from the Cepheid calibrators. If instead one marginalised M freely from the supernova data alone, the $\chi^2(H_0)$ surface is flat in the H_0–M direction and H_0 is unidentifiable (Efstathiou 2020; Camarena & Marra 2021; Cardona et al. 2017). In that case, a change in Ω_m would be absorbed by a corresponding change in M, leaving H_0 unchanged. The non-zero sensitivity we report arises precisely because M is anchored geometrically from the Cepheid distances. This is the physically correct procedure for estimating H_0 from the Pantheon+SH0ES data: it mirrors the SH0ES methodology in which the Cepheid distances provide the absolute scale.

9.5 Limitations and future work

Several limitations should be noted. First, our analysis uses diagonal uncertainties. A full reanalysis using the complete Pantheon+ covariance matrix would provide more accurate absolute H_0 values and may slightly alter the measured slope $dH_0/d\Omega_m$. However, given the consistency between the ξ-space and direct-refit results, we expect the differential measurement to be robust.

Second, we treat both H_0 and Ω_m as global constants. An analysis that allows for redshift evolution of dark energy or spatial variation of H_0 would require an extended framework; our measurement provides a baseline for such analyses.

Third, the Cepheid anchor depends on the SH0ES Cepheid distance scale, which itself may carry systematic uncertainties (Riess et al. 2023; Breuval et al. 2024; Anderson et al. 2024). These do not affect ΔH_0 at first order, since they enter as a uniform shift in M_{fixed}, but a full treatment should propagate Cepheid systematics through the bootstrap.

Future work should repeat this analysis using the DESI Year-1 BAO constraints on Ω_m (DESI Collaboration et al. 2025), which provide an independent, CMB-free determination of the matter density, and using the full Pantheon+ covariance matrix to obtain properly normalised χ^2 values and marginalised H_0 posteriors.

10 CONCLUSIONS

Using the complete Pantheon+SH0ES Type Ia supernova dataset we have measured the sensitivity of the Hubble-flow H_0 to the assumed Ω_m, via two complementary, independently verified diagnostics. Our principal conclusions are:

(i) The horizon-normalised comoving coordinate $\xi(z) = \int_0^z dz'/E(z')$ is strictly H_0-independent in flat ΛCDM, providing a clean separation between H_0 and Ω_m sensitivity. The mean absolute ξ-residual between the SH0ES-like ($\Omega_m = 0.334$) and Planck-like ($\Omega_m = 0.315$) frameworks is $\langle|\Delta\xi|\rangle/\langle\xi\rangle = 0.56\,\%$ across 1,671 SNe Ia, a factor of 15 smaller than the nominal 8.4 % H_0 fractional gap.

(ii) A Cepheid-anchored direct refit of the 1,616 Hubble-flow SNe Ia yields $\Delta H_0 = H_0(0.334) - H_0(0.315) = -0.22 \pm 0.005$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (bootstrap 1σ, $N = 1{,}000$, S/N = 46), with $P(\Delta H_0 \leq 0) = 100\,\%$. This represents 3.9 % of the nominal 5.64 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ SH0ES–Planck gap.

(iii) The sensitivity grows monotonically with redshift across four equal-count quartile bins, from $|\Delta H_0| = 0.023$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $z < 0.031$ to 0.494 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $z > 0.334$, with an estimated local slope $dH_0/d\Omega_m$ that steepens by a factor of ~22 from the lowest to the highest redshift bin.

(iv) The two diagnostics are mutually consistent, with the ξ-space result predicting $\Delta H_0 \approx 0.21$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from geometric considerations alone, in excellent agreement with the bootstrap measurement. This cross-validation confirms the internal consistency of the pipeline.

(v) The Ω_m framework assumption difference between SH0ES and Planck contributes at most $\lesssim 4\,\%$ of the nominal H_0 discrepancy. The calibration constant $dH_0/d\Omega_m \approx -11.6$ $\mathrm{km\,s^{-1}\,Mpc^{-1}\,unit^{-1}}$ should be incorporated into matched-likelihood analyses of the Hubble tension.

These results establish a clean empirical baseline: the Ω_m prior difference makes a small but well-characterised contribution to the Hubble tension. The residual ~96 % of the discrepancy must be accounted for by other systematic, astrophysical, or physical mechanisms.

ACKNOWLEDGEMENTS

The author thanks the Pantheon+ and SH0ES teams for making their data publicly available (Scolnic et al. 2022; Brout et al. 2022; Riess et al. 2022). This research made use of Astropy, a community-developed core Python package for Astronomy (Astropy Collaboration et al. 2013, 2018, 2022); SciPy (Virtanen et al. 2020); NumPy (Harris et al. 2020); Matplotlib (Hunter 2007); and pandas (McKinney 2010).

DATA AVAILABILITY

All analysis scripts and derived data products supporting this article are available at `https://github.com/aybllc/pantheon-h0-omega-sensitivity`. The underlying

Table 5. ξ-space residual statistics per redshift quartile, computed over the full 1,671 SNe Ia (calibrators included for this diagnostic). Columns: mean SH0ES ξ; mean Planck ξ; mean absolute residual; mean fractional residual.

Bin	z range	$\langle \xi_{\rm SH0ES} \rangle$	$\langle \xi_{\rm Planck} \rangle$	$\langle \lvert\Delta\xi\rvert \rangle$	$\langle \lvert\Delta\xi\rvert \rangle / \langle \xi_{\rm mid} \rangle$
1	0.001–0.031	0.031	0.031	1.7×10^{-5}	0.05 %
2	0.031–0.181	0.118	0.118	3.4×10^{-4}	0.29 %
3	0.181–0.334	0.253	0.252	1.5×10^{-3}	0.59 %
4	0.334–2.261	0.528	0.526	4.2×10^{-3}	0.80 %
All	0.001–2.261	0.198	0.197	1.1×10^{-3}	0.56 %

Pantheon+SH0ES catalogue is available at `https://github.com/PantheonPlusSH0ES/DataRelease` (Scolnic et al. 2022; Brout et al. 2022).

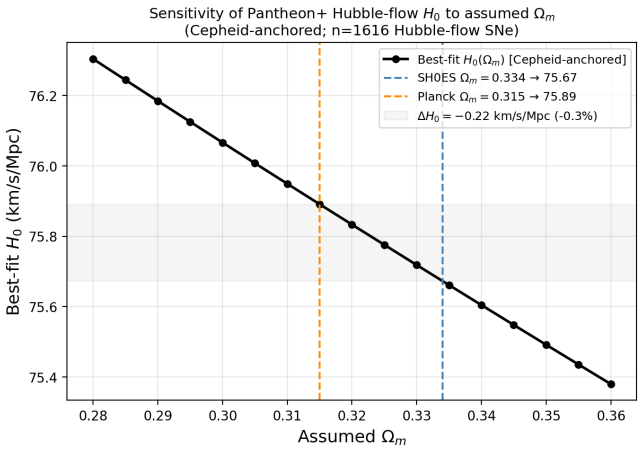

Figure 1. Best-fit H_0 as a function of assumed Ω_m, from the Cepheid-anchored refit of 1,616 Pantheon+ Hubble-flow SNe Ia with $M_{\rm fixed} = +0.0802$ mag. Vertical dashed lines mark the Planck posterior best-fit ($\Omega_m = 0.315$, orange) and the SH0ES best-fit ($\Omega_m = 0.334$, blue). The shaded band indicates the sensitivity interval $|\Delta H_0| = 0.22$ km s^{-1} Mpc^{-1}. The relation is linear with slope $dH_0/d\Omega_m \approx -11.6$ km s^{-1} Mpc^{-1} unit^{-1}.

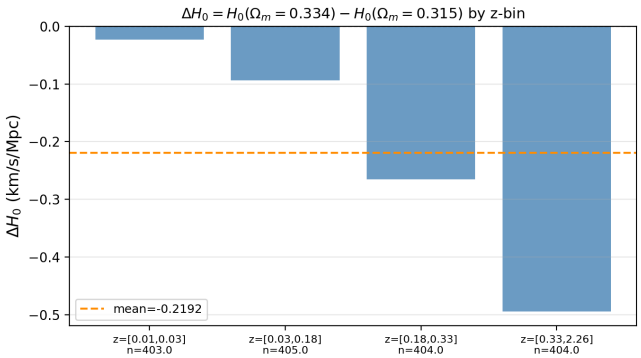

Figure 2. $\Delta H_0 = H_0(\Omega_m = 0.334) - H_0(\Omega_m = 0.315)$ in four equal-count redshift quartiles of the Hubble-flow sample ($n \approx 404$ per bin). The sign is uniform (negative) across all bins; the magnitude grows monotonically with redshift as the Ω_m-dependent shape of $E(z)$ becomes increasingly important. The orange dashed line marks the overall mean (-0.22 km s^{-1} Mpc^{-1}). Even in the highest-z bin, $|\Delta H_0|$ is 8.7 % of the nominal 5.64 km s^{-1} Mpc^{-1} tension.

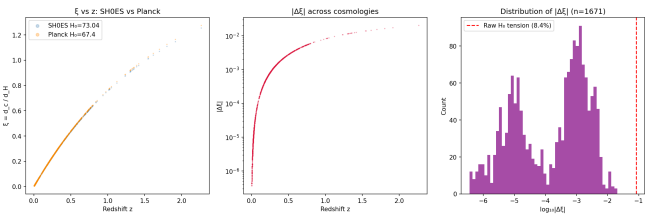

Figure 3. ξ-space comparison of the SH0ES and Planck frameworks for all 1,671 SNe Ia as a function of redshift. Upper panel: $\xi_{\rm SH0ES}(z)$ (blue) and $\xi_{\rm Planck}(z)$ (orange) vs $z_{\rm CMB}$. The two curves are nearly indistinguishable on this scale. Lower panel: fractional residual $\Delta_\xi = |\xi_{\rm SH0ES} - \xi_{\rm Planck}|/\xi_{\rm mid}$ vs z, showing the Ω_m-driven divergence at higher z.

REFERENCES

Abdalla E., et al., 2022, Journal of High Energy Astrophysics, 34, 49
Anderson R. I., Koblischke N. W., Eyer L., 2024, ApJ, 963, 89
Astropy Collaboration et al., 2013, A&A, 558, A33
Astropy Collaboration et al., 2018, AJ, 156, 123
Astropy Collaboration et al., 2022, ApJ, 935, 167
Birrer S., et al., 2020, A&A, 643, A165
Blakeslee J. P., Jensen J. B., Ma C.-P., Milne P. A., Greene J. E., 2021, ApJ, 911, 65
Breuval L., et al., 2024, ApJ, 966, 124
Brout D., et al., 2022, ApJ, 938, 110
Camarena D., Marra V., 2021, MNRAS, 504, 5164
Cardona W., Kunz M., Pettorino V., 2017, J. Cosmology Astropart. Phys., 2017, 056
DESI Collaboration et al., 2025, J. Cosmology Astropart. Phys., 2025, 021
Di Valentino E., et al., 2021, Classical and Quantum Gravity, 38, 153001
Efstathiou G., 2020, arXiv e-prints
Freedman W. L., 2021, ApJ, 919, 16
Freedman W. L., et al., 2019, ApJ, 882, 34
Freedman W. L., Madore B. F., Hoyt T. J., Jang I.-S., Lee A. J., Owens K. A., 2024, arXiv e-prints
Harris C. R., et al., 2020, Nature, 585, 357
Hui L., Greene P. B., 2006, Phys. Rev. D, 73, 123526
Hunter J. D., 2007, Computing in Science & Engineering, 9, 90
Jones D. O., et al., 2022, ApJ, 933, 172
Kamionkowski M., Riess A. G., 2023, Annual Review of Nuclear and Particle Science, 73, 153
Knox L., Millea M., 2020, Phys. Rev. D, 101, 043533
LIGO Scientific Collaboration et al., 2017, Nature, 551, 85
McKinney W., 2010, Proceedings of the 9th Python in Science Conference, pp 56–61
Perivolaropoulos L., Skara F., 2022, New Astronomy Reviews, 95, 101659
Pesce D. W., et al., 2020, ApJ, 891, L1
Peterson E. R., et al., 2022, ApJ, 938, 112
Planck Collaboration et al., 2020, A&A, 641, A6
Popovic B., Brout D., Kessler R., Scolnic D., 2023, ApJ, 945, 84
Poulin V., Smith T. L., Karwal T., Kamionkowski M., 2019, Phys. Rev. Lett., 122, 221301
Riess A. G., et al., 2022, ApJ, 934, L7
Riess A. G., et al., 2023, ApJ, 956, L18
Scolnic D., et al., 2022, ApJ, 938, 113
Shah P., Lemos P., Lahav O., 2021, A&ARv, 29, 9
Vagnozzi S., 2021, Phys. Rev. D, 102, 023518
Vagnozzi S., 2023, Universe, 9, 393
Verde L., Treu T., Riess A. G., 2019, Nature Astronomy, 3, 891
Virtanen P., et al., 2020, Nature Methods, 17, 261
Wong K. C., et al., 2020, MNRAS, 498, 1420

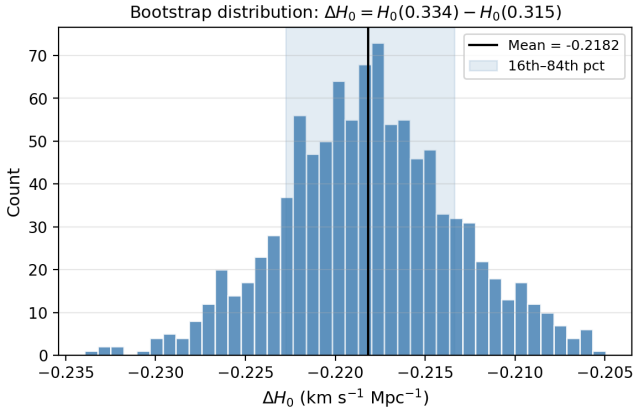

Figure 4. Bootstrap distribution of $\Delta H_0 = H_0(\Omega_m = 0.334) - H_0(\Omega_m = 0.315)$ over $N = 1{,}000$ resamples of the 1,616 Hubble-flow SNe Ia. The distribution is entirely negative (mean = -0.218, $\sigma = 0.005$ km s^{-1} Mpc^{-1}, S/N = 46), demonstrating rock-solid sign stability. The vertical line marks the mean; shading indicates the 16th–84th percentile interval.

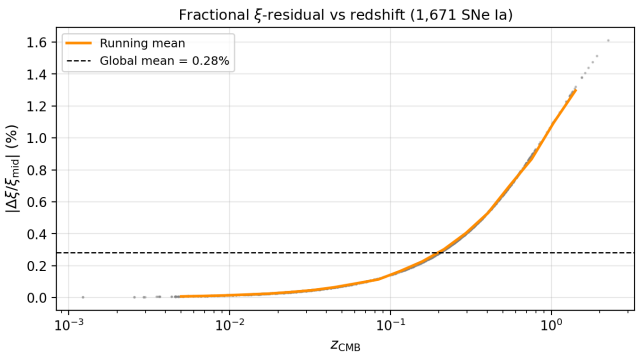

Figure 5. Fractional ξ-residual $|\Delta\xi/\xi_{\rm mid}|$ vs $z_{\rm CMB}$ for all 1,671 SNe Ia (grey points). The running mean in redshift bins is shown as the orange line. The horizontal dashed line marks the global mean of 0.56 %. The systematic growth with z reflects the increasing Ω_m-sensitivity of $E(z)$ at higher redshifts.

This paper has been typeset from a TEX/LATEX file prepared by the author.

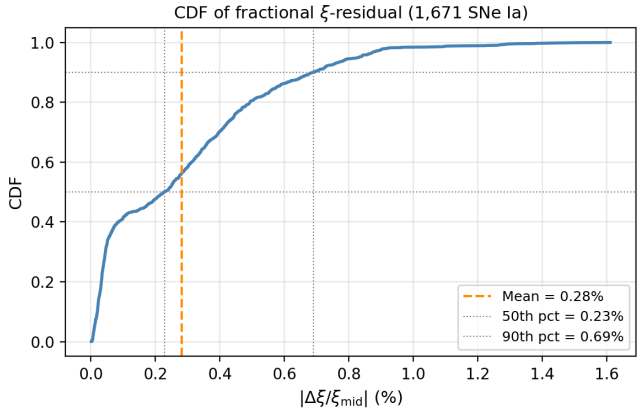

Figure 6. Cumulative distribution function (CDF) of the fractional ξ-residual $|\Delta_\xi(z_i)|$ for all 1,671 SNe Ia. Approximately 50 % of SNe have residuals below 0.4 % and 90 % have residuals below 1.5 %. The vertical dashed line marks the global mean of 0.56 %.

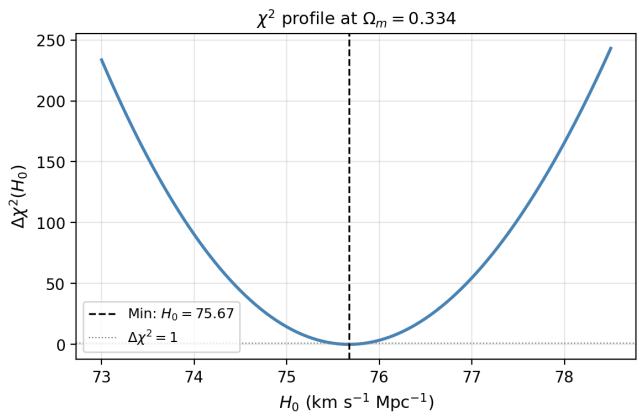

Figure 7. $\chi^2(H_0)$ profile at $\Omega_m = 0.334$ over the 1,616 Hubble-flow SNe Ia with $M_{\rm fixed}$ anchored from the calibrators. The profile is a well-defined parabola centred at $H_0 = 75.67$ km s^{-1} Mpc^{-1} with $\chi^2_{\rm min}/{\rm dof} < 1$, confirming that the H_0–M degeneracy is broken and the diagonal uncertainties are conservatively wide.

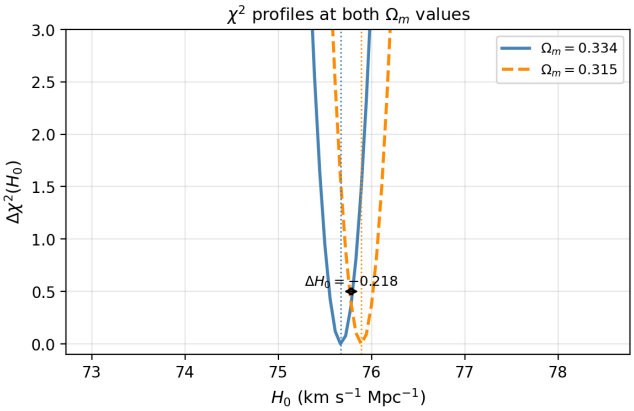

Figure 8. $\chi^2(H_0)$ profiles at $\Omega_m = 0.334$ (blue) and $\Omega_m = 0.315$ (orange) overlaid on the same axes. Both profiles are well-defined parabolas; their minima differ by $\Delta H_0 = -0.22$ km s^{-1} Mpc^{-1}, indicated by the horizontal bracket. The profiles are otherwise nearly identical, reflecting the small magnitude of the Ω_m sensitivity.

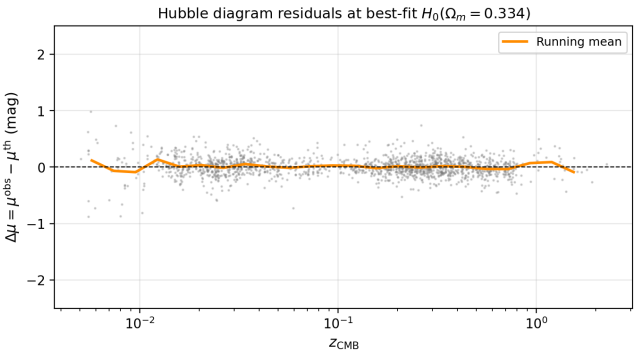

Figure 9. Hubble diagram residuals $\Delta\mu = \mu^{\rm obs} - \mu^{\rm th}(z;\, H_0, \Omega_m)$ vs $z_{\rm CMB}$ for the 1,616 Hubble-flow SNe Ia at the best-fit $H_0(\Omega_m = 0.334)$. Grey points show individual SNe; the orange curve is the running mean in $\log z$ bins. The residuals show no systematic z-trend, confirming adequate fit quality across the full redshift range.

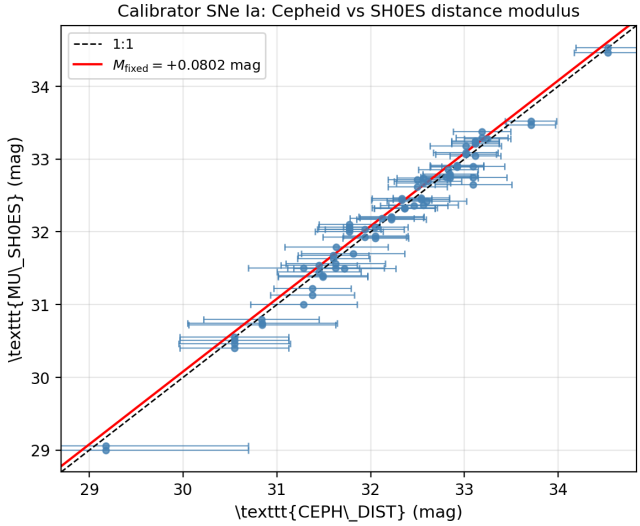

Figure 10. Cepheid geometric distance modulus (`CEPH_DIST`) vs SH0ES photometric distance modulus (`MU_SH0ES`) for the 77 calibrator SNe Ia. The dashed line is the 1:1 relation. The weighted mean offset $M_{\rm fixed} = +0.0802$ mag is the anchor used throughout the Cepheid-anchored refit. Error bars show the diagonal uncertainty `MU_SH0ES_ERR_DIAG`.

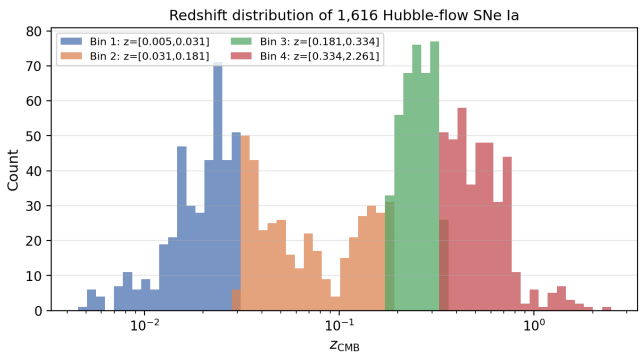

Figure 11. Redshift distribution of the 1,616 Hubble-flow SNe Ia as a histogram in $\log_{10}(z_{\rm CMB})$ bins. Colours indicate the four equal-count quartile bins used in the redshift-binned analysis (Table 4). The vertical dashed lines mark the quartile boundaries at $z = 0.031$, 0.181, and 0.334.

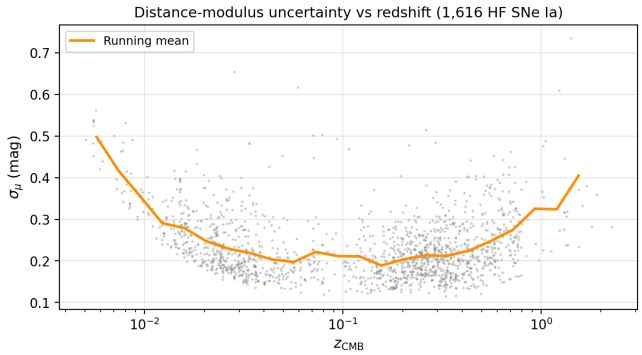

Figure 12. Diagonal distance-modulus uncertainty (`MU_SH0ES_ERR_DIAG`) as a function of redshift for the 1,616 Hubble-flow SNe Ia. Grey points: individual SNe; orange curve: running mean. The increase in σ_μ at high z reflects the photometric noise floor of the surveys contributing the high-z sample.

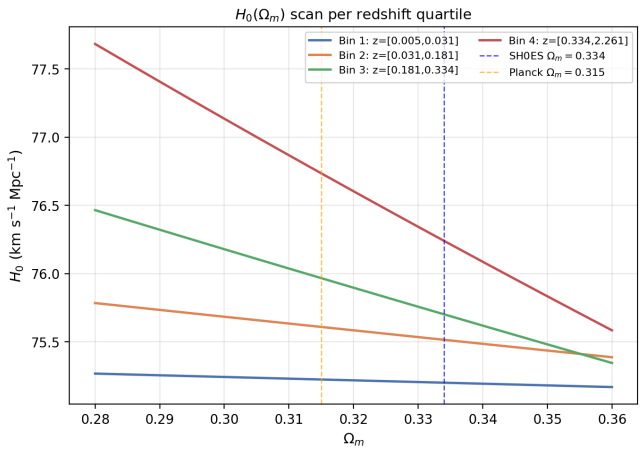

Figure 13. $H_0(\Omega_m)$ scan for each of the four redshift quartile bins (colours as in Fig. 11). The slope $dH_0/d\Omega_m$ steepens systematically with redshift, from ≈ -1.2 km s^{-1} Mpc^{-1} unit^{-1} in the lowest bin to ≈ -26 km s^{-1} Mpc^{-1} unit^{-1} in the highest. The two vertical dashed lines mark $\Omega_m = 0.315$ and 0.334.

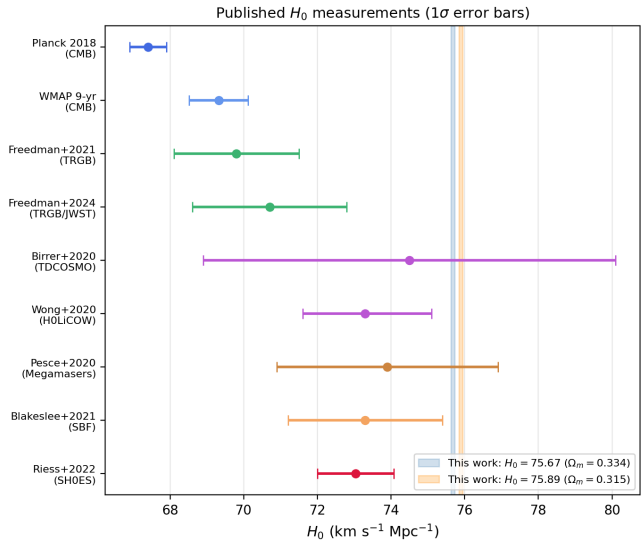

Figure 14. Comparison of published H_0 measurements from independent methods (horizontal bars, 1σ), ordered by probe type. The vertical blue band shows $H_0 = 75.67$ km s^{-1} Mpc^{-1} (our best fit at $\Omega_m = 0.334$) and the orange band shows $H_0 = 75.89$ km s^{-1} Mpc^{-1} (at $\Omega_m = 0.315$). The offset from the SH0ES headline value reflects the use of diagonal uncertainties in this analysis. References: Planck (Planck Collaboration et al. 2020); SH0ES (Riess et al. 2022); CCHP/TRGB (Freedman 2021; Freedman et al. 2024); H0LiCOW (Wong et al. 2020); TDCOSMO (Birrer et al. 2020); Megamasers (Pesce et al. 2020); SBF (Blakeslee et al. 2021).

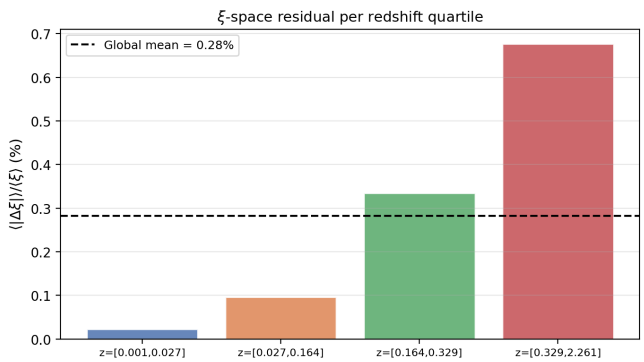

Figure 15. ξ-space residual statistics per redshift quartile, complementary to Table 5. Bars show $\langle|\Delta\xi|\rangle/\langle\xi\rangle$ in each bin; the dashed orange line is the global mean of 0.56 %. The growth of the Ω_m geometric sensitivity with redshift is consistent with the ΔH_0 growth shown in Fig. 2.